

Information Management

Assignment 5

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Q1. E-R Diagrams as graphs

Disconnected Graph is an indicator that the enterprise schema contains two or more sets of entities and relationships that have no functional connection to each other. In a practical sense, it means the database is storing data for multiple independent systems that do not share any common associations.

Graph with a Cycle: means that a cycle suggests that there are multiple paths or relationships connecting a set of entities. This represents redundant relationships where an entity can be reached from itself via a chain of other entities.

Q3 why do we not change weak entity sets?

1. Weak entity sets prevent the need to replicate the primary keys of strong entities within the weak entity itself, which keeps the schema cleaner. This avoids redundancy
 2. By using a weak entity, the database can automatically handle problems with multiple deleting of tuples ensuring that if a strong entity is removed, all associated dependent records are also purged thus again keeping it all clean within the schema.
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Q4a.

1. `SELECT e.ID, e.person_name FROM employee AS e, works AS w, company AS c WHERE e.ID = w.ID AND w.company_name = c.company_name AND e.city = c.city`
2. `SELECT e.ID, e.person_name FROM employee AS e, manages AS m, employee AS manager WHERE e.ID = m.ID AND m.manager_id = manager.ID AND e.street = manager.street AND e.city = manager.city`
3. `SELECT e.ID, e.person_name FROM employee AS e, works AS w WHERE e.ID = w.ID AND w.salary > ( SELECT AVG(w2.salary) FROM works AS w2 WHERE w2.company_name = w.company_name )`

Q4b In Natural joins it automatically join tables based on *all* columns that share the same name. The query would incorrectly filter for instructors who are teaching courses only within their *own* department, failing to list instructors who are teaching classes in other departments.
